

XII. Real-Robot Experiments

Real-robot experiments are conducted to validate the executability of the proposed planning-aware LTL-MRTA framework in a warehouse environment and to evaluate online mission repair when a robot becomes unavailable during execution. The experiments use four mobile robots, $\mathcal{R} = \{r_1, r_2, r_3, r_4\}$, belonging to two robot types. Robots r_1 and r_2 are type-1 robots, whereas r_3 and r_4 are type-2 robots. Two experimental conditions are considered: a nominal mission with all four robots available and a robot-unavailability mission in which r_3 is deliberately removed from the active team during task execution.

A. Experimental Setup and Warehouse Mission

The experimental workspace contains five regions of interest $\mathcal{L} = \{l_1, l_2, l_3, l_4, l_5\}$. Regions l_1 and l_4 represent the inbound and outbound areas, respectively; l_3 is the central control area; and l_2 and l_5 are shelf areas. To maintain consistency with the task model in Section III, the physical operations are expressed using action propositions a_i rather than region propositions p_i . Each action a_i is associated with an execution requirement $c(a_i) = (\tau_i, n_i, l_i)$, where τ_i is the required robot type, n_i is the required number of robots, and l_i is the execution region. All service durations are set to zero in the experiment; therefore, an action is marked complete once the required robots reach their designated target points and the corresponding completion condition is confirmed.

The warehouse mission is specified by

$$\varphi_{\text{hw}} = \Diamond a_1 \wedge \Diamond(a_2 \wedge (\Diamond a_3 \wedge \Diamond a_4)) \wedge \Diamond(a_5 \wedge a_6). \quad (1)$$

The action requirements and target points used by the low-level controller are listed in Table I. In application terms, a_1 represents an operation in the inbound area, a_2 a central-control inspection, a_3 a two-robot collaborative shelf inspection, a_4 a shelf-area inspection, and a_5 – a_6 a heterogeneous outbound verification. These labels provide warehouse semantics without assigning robot capabilities beyond the two abstract types used by the planner.

TABLE I: Action propositions used in the real-robot warehouse experiment.

Action requirement	Target point(s)	Warehouse interpretation
$c(a_1) = (1, 1, l_1)$	(239, 241)	Inbound-area inspection / confirmation
$c(a_2) = (2, 1, l_3)$	(250, 197)	Central-control inspection / status confirmation
$c(a_3) = (1, 2, l_2)$	(236, 161), (254, 159)	Collaborative shelf inspection requiring both type-1 robots
$c(a_4) = (1, 1, l_5)$	(313, 165)	Shelf-area inspection
$c(a_5) = (1, 1, l_4)$	(306, 237)	Type-1 outbound-area verification
$c(a_6) = (2, 1, l_4)$	(324, 235)	Type-2 outbound-area verification

The formula in (1) contains several task structures that are relevant to warehouse inspection. First, a_1 is an independent eventual operation. Second, a_2 acts as a temporal gate: after a_2 is satisfied, both a_3 and a_4 must eventually be completed, while no fixed order is imposed between a_3 and a_4 . Third, a_3 requires simultaneous participation of two robots of the

same type. Because r_1 and r_2 are the only type-1 robots, both must participate in a_3 . Finally, $\Diamond(a_5 \wedge a_6)$ requires a type-1 robot and a type-2 robot to satisfy the two outbound conditions jointly in l_4 . Thus, the same experiment evaluates temporal precedence, homogeneous collaboration, heterogeneous synchronization, and online reassignment in one compact mission.

B. Nominal Mission Execution

We first execute φ_{hw} with all four robots available. The proposed framework generates a task-level feasible plan and dispatches type-compatible robots to the corresponding target points. For the single-robot type-1 actions a_1 , a_4 , and a_5 , either r_1 or r_2 can be selected according to the execution context retained by the planner. Similarly, either r_3 or r_4 can execute the single-robot type-2 actions a_2 and a_6 . In contrast, the collaborative action a_3 necessarily uses both r_1 and r_2 , which are directed to the two target points in l_2 and the action becomes true only after both robots satisfy the completion condition.

The final outbound operation explicitly tests heterogeneous synchronization. One type-1 robot is assigned to the target of a_5 and one type-2 robot is assigned to the target of a_6 in l_4 . The conjunction $a_5 \wedge a_6$ is accepted only after both conditions are satisfied, rather than when either robot reaches the outbound area individually. The nominal run therefore provides an end-to-end validation that the task-level plan, type constraints, multi-robot synchronization, and physical execution are consistent with the LTL specification.

C. Online Repair Under Robot Unavailability

The second experiment evaluates mission continuity when the available team changes during execution. Robot r_3 is deliberately made unavailable while the mission is active. After the failure event, the active robot set becomes

$$\mathcal{R}' = \{r_1, r_2, r_4\}. \quad (2)$$

Once the unavailability of r_3 is detected, assignments involving r_3 that have not yet been completed are invalidated. Confirmed completed actions are retained, and replanning is initialized from the currently monitored automaton progress together with the current states of the surviving robots. No subsequent action is assigned to r_3 .

The mission remains feasible after the failure because each type-2 action, a_2 and a_6 , requires only one type-2 robot. The surviving type-2 robot r_4 can therefore take over any unfinished type-2 operation that was previously assigned to r_3 . The type-1 portion of the mission remains unchanged: r_1 and r_2 continue to execute the single-robot type-1 actions and must still cooperate for a_3 . In particular, if the failure occurs before the completion of a_2 or a_6 , the repaired plan redirects r_4 to the corresponding unfinished target while preserving all task progress that has already been confirmed.

The repaired run completes the remaining mission from the current execution state without restarting from the initial condition, validating physical executability and online repair under robot unavailability.

编辑说明（不属于论文正文）

上面三小节按你的论文符号体系和参考文献的实验叙事方式整理：先交代实物平台/区域/机器人类型，再给出 LTL 任务和 action requirements，随后分别写正常执行与机器人故障恢复。参考论文的实物实验也是按“场景与原子任务定义 → 正常任务 → 故障/变化后的重规划 → 结果解释”的方式展开。

1. 已严格按你提供的信息写入的内容

- 任务统一改为 action proposition: $\varphi_{hw} = \Diamond a_1 \wedge \Diamond(a_2 \wedge (\Diamond a_3 \wedge \Diamond a_4)) \wedge \Diamond(a_5 \wedge a_6)$ 。
- 五个区域的业务语义: l_1 入库区、 l_4 出库区、 l_3 中央控制区、 l_2/l_5 货架区。
- 四台机器人: r_1, r_2 为 type 1, r_3, r_4 为 type 2。
- 六个动作的机器人类型、数量、区域以及全部目标点坐标。
- 两组实物实验: 无故障正常执行, 以及 r_3 故障后的在线修复。
- 没有把未知的规划时间、故障时刻、机器人型号或重复次数编造成实验结果。

2. 正式投稿前建议补充的实测信息

为了让这一节达到参考论文的完整度, 建议在有数据后补充以下信息; 这些内容目前没有提供, 所以正文中没有虚构:

- 1) 机器人具体平台、定位方式、低层路径规划/避障方法, 以及高层规划运行的计算机配置;
- 2) 正常实验的高层首个可行计划时间 T_{first} , 如有 BnB 优化再给 T_{best} ;
- 3) r_3 的实际故障触发时刻, 以及从故障检测到获得修复计划的 replanning latency;
- 4) 每组实物实验的重复次数和成功率, 最好至少 5 次, 并给均值/标准差或 median/IQR;
- 5) 两组实验的任务总完成时间, 便于量化故障引入的 mission delay;
- 6) 如果你保存了实际机器人分配序列, 可在正文中像参考论文一样明确写出具体哪台机器人执行了 a_1, a_2, a_4, a_5, a_6 。

3. 建议配套的三张图及图注

Fig. X. Experimental environment and task abstraction. (a) Real-robot warehouse environment. (b) Abstract map with five regions of interest. Regions l_1 and l_4 denote the inbound and outbound areas, l_3 denotes the central control area, and l_2 and l_5 denote shelf areas. The marked points correspond to the execution targets of a_1 – a_6 .

Fig. Y. Nominal execution of the warehouse mission. Representative snapshots show the execution of the LTL mission with all four robots available. The two type-1 robots cooperate for a_3 in l_2 , while the final outbound operation requires one type-1 robot and one type-2 robot to jointly satisfy $a_5 \wedge a_6$ in l_4 .

Fig. Z. Online repair after robot unavailability. (a) Robot r_3 becomes unavailable during mission execution and replanning is triggered. (b) The unfinished assignments involving r_3 are invalidated and the surviving type-2 robot r_4 is reassigned to the remaining type-2 action. (c) The surviving team completes the remaining mission without assigning subsequent tasks to r_3 .

4. 一个需要注意的 LTL 语义点

公式 $\Diamond(a_2 \wedge (\Diamond a_3 \wedge \Diamond a_4))$ 只要求 a_3 和 a_4 都发生在某次满足 a_2 之后, 并没有规定 a_3 与 a_4 谁先谁后。如果你的真实任务需要严格顺序“ a_2 后先 a_3 再 a_4 ”, 建议把公式改为类似 $\Diamond(a_2 \wedge \Diamond(a_3 \wedge \Diamond a_4))$, 再重新核对自动机与实物实验。